Andrés Nahuel Páez

Laboratory Software Developer | Data · Quality · Traceability

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PROFILE

Software and data developer focused on laboratories. I build tools with deterministic calculations verified against published references (NIST StRD, EURACHEM/GUM, Westgard rules), hash-chained audit trails, electronic signatures with re-authentication, and instrument integration via ASTM E1394 and HL7 v2. I use AI (Claude API) only off the critical path, with human review and logging. I bring 8 years of administrative and records management at educational institutions (sensitive files, deadlines and regulations), plus training in statistics, geology and hydrology.

Availability: remote worldwide (UTC−3: overlaps the European afternoon and most of the working day in the Americas). Open to relocation. Immediate start.

PROJECTS

Laboratory software portfolio — Next.js · TypeScript · PostgreSQL · Claude API

`andresnahuelpaez-portfolio-laboratorios.vercel.app`

- Integrated flow Enlace → Quality control → Custodia → Report: ASTM E1394 / HL7 v2 parser with LOINC mapping, Westgard multirules that block release, SHA-256 audit trail and electronic signatures with segregation of duties.
- Calibration (weighted least squares, LOD/LOQ, inverse prediction with 95 % CI) verified against NIST StRD Norris; GUM + Monte Carlo uncertainty verified against EURACHEM A1.
- GAMP 5-style validation dossier: URS, risk assessment and a traceability matrix generated automatically from the tests (Vitest + Playwright, CI on GitHub Actions).
- PostgreSQL schema with a trigger-enforced append-only audit trail, tested with PGlite; Claude API assistant under an AI use policy (draft EU GMP Annex 22; the public demo runs in rule-based offline mode).

NormaMina: RAG assistant over mining regulations — Next.js · Supabase pgvector · Google Gemini

`normamina.vercel.app`

- Answers cite the document and article; when there is no information, it says so.

Solar generation forecasting with machine learning — Python · scikit-learn

- Random Forest with R^2 0.977 and −70 % error vs. baseline, on synthetic data built from real physical parameters.

Lithium brine evaporation simulator — Python

- Brine balance across a cascade of ponds (Mg/Li, precipitated salts, LCE).

REGULATORY KNOWLEDGE

FDA 21 CFR Part 11 · EU GMP Annex 11 and draft Annex 22 (AI) · GAMP 5 · ALCOA+ · ISO/IEC 17025 · ISO 15189 — applied in personal projects.

TECHNICAL SKILLS

- Languages and data: Python (pandas, NumPy, scikit-learn, matplotlib), SQL (PostgreSQL, SQLite, Supabase), TypeScript.
- Web and software quality: Next.js, React, Tailwind, Vitest, Playwright, Git, GitHub Actions, Vercel.
- Laboratory and integration: ASTM E1394 / LIS2-A2, HL7 v2 (ORU^R01), LOINC, Westgard, SPC, calibration, GUM uncertainty, LIMS/LIS (demo core), data integrity (ALCOA+), computer system validation (CSV, GAMP 5, GxP).
- AI: Claude API, RAG with citations, Model Context Protocol (MCP), text-to-SQL, AI with GxP controls (draft, human review, logging).
- GIS: PostGIS, GeoPandas, MapLibre, Leaflet.

EXPERIENCE

Independent software and data developer — La Rioja, Argentina (remote) · 2025 – present

- Built a laboratory software portfolio, NormaMina (RAG over mining regulations) and personal data and machine-learning projects in mining, energy and finance.
- Next.js/TypeScript, Python, SQL and Claude API, with automated tests, continuous integration and deployment on Vercel.

Administrative Secretary — Colegio Pbro. Virgilio Carlos Ferreyra, Anillaco (La Rioja) · 2017 – 2025

- Document and records management involving sensitive information, deadlines and regulations.
- Coordination of administrative processes and operational planning.

Academic and Administrative Coordinator — ISFD Madre Teresa de Calcuta, Anjullón (La Rioja) · 2019 – 2025

- Administrative follow-up, team coordination and process improvement.

Secondary School Teacher — Schools in La Rioja · 2015 – 2019

- Lesson planning, teaching and assessment.

EDUCATION

Teaching Degree in Geography — I.S.F.D. Albino G. Sánchez Barros, La Rioja · Completed 2014 · issued 2016

- Applied statistics and computing, cartography and photogrammetry, geology and geomorphology, climatology and hydrology, quantitative research.

Bachelor's Degree in Education — Universidad Siglo 21 · 2017

- Institutional quality management, research methodology, project planning.

English Language Technician — Instituto English School, La Rioja · 2015

- English Language Technician program (2012–2014); intermediate level.

CERTIFICATIONS

- Anthropic: Claude Code in Action (2026)
- Anthropic: Claude with the Anthropic API (2026)
- Anthropic: Introduction to Model Context Protocol (2026)
- Anthropic: AI Fluency: Framework & Foundations (2026)
- Fundación YPF: Industrial Instrumentation (16 h) (2026)
- Fundación YPF: Occupational Health and Safety (16 h) (2026)
- Fundación YPF: Renewable Energy (16 h) (2026)
- Fundación YPF: Introduction to Artificial Intelligence (3 h) (2026)
- Fundación YPF: Excel (5 h) (2026)

WHAT I CAN BUILD

- Migrate Excel spreadsheets to a system with audit trail, roles and signatures.
- Automatic reports and certificates of analysis from audited data.
- Ingest instrument files and interfaces (CSV, ASTM, HL7) into a database.
- AI assistants over SOPs and regulations, with source citations and human review.
- Validation documentation: URS, risk assessment, traceability matrix and protocols.

LANGUAGES

Spanish: native · English: intermediate (English Language Technician, 2015).